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(54) **GAME BALL WITH VARIABLE LOBULAR SHAPE**

Publication Classification

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(52) **U.S. Cl.**
CPC **A63B 41/08** (2013.01)

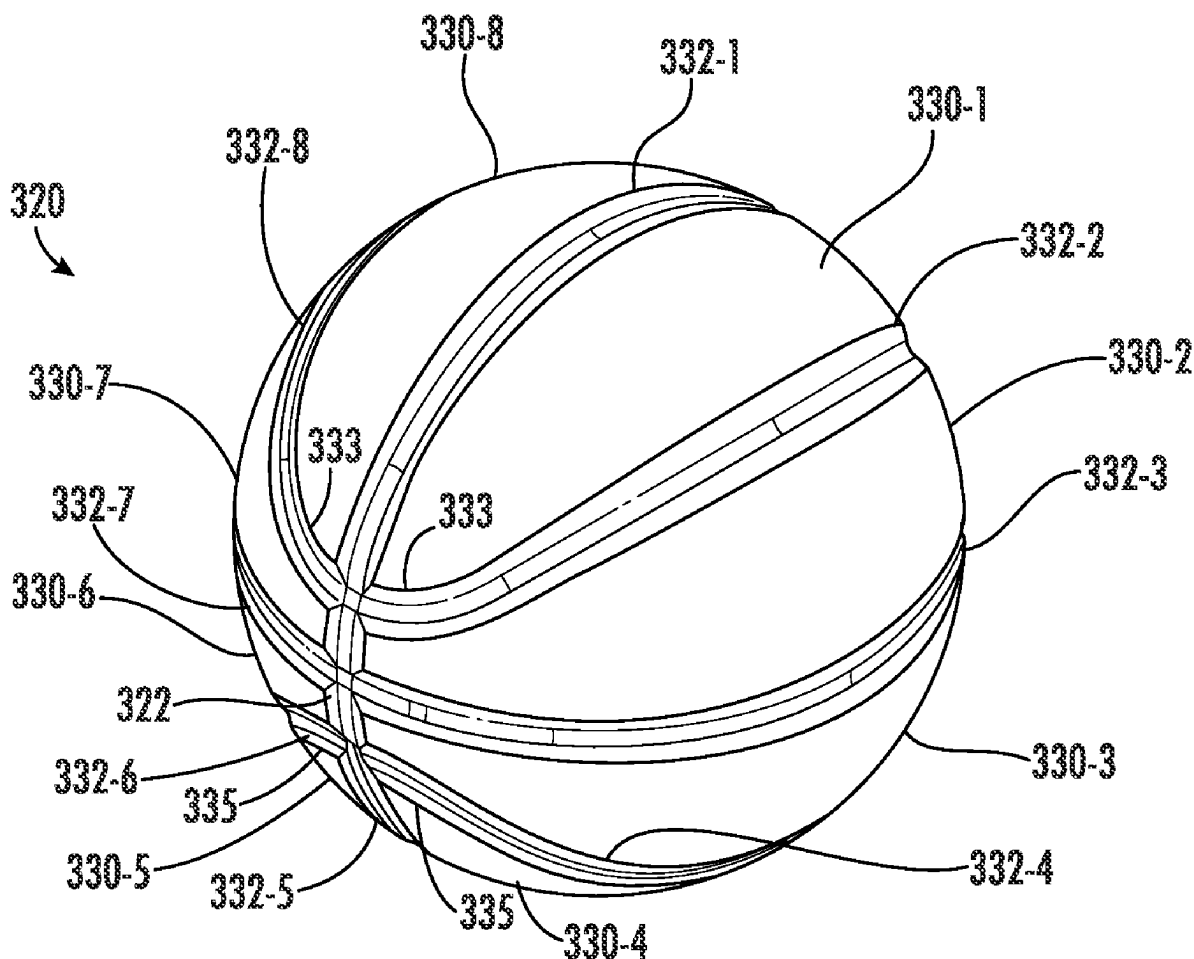
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(57) **ABSTRACT**

(21) Appl. No.: **18/592,021**

A variable loby game ball has an outer surface that includes a first pole, a second pole opposite the first pole and outer lobes. The outer lobes extend between the first pole and the second pole. Recessed channels that vary in at least one of width, depth and cross-sectional shape extend along consecutive lobes.

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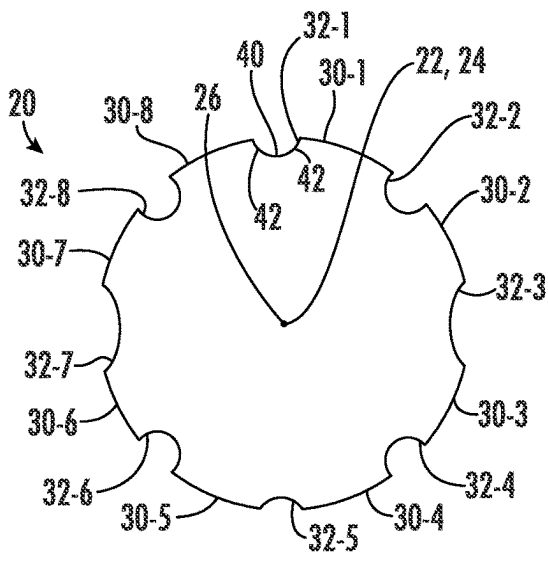


FIG. 1

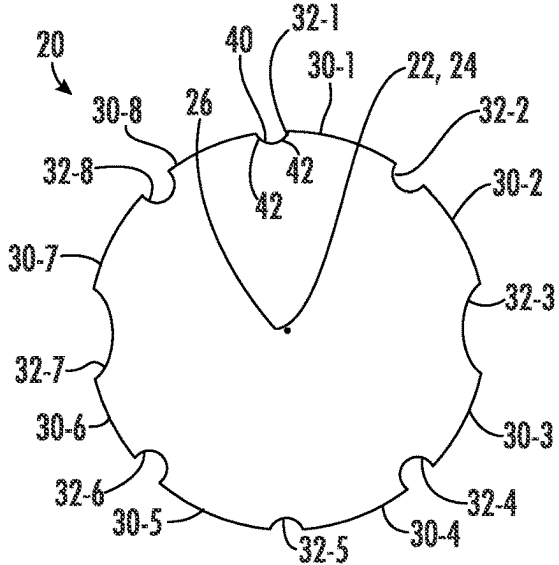


FIG. 2

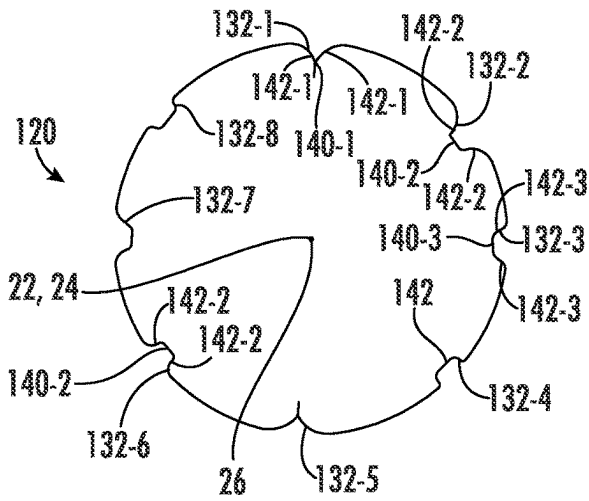


FIG. 3

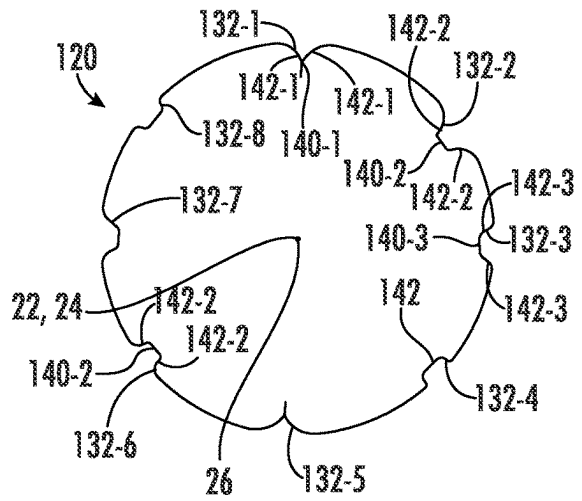


FIG. 4

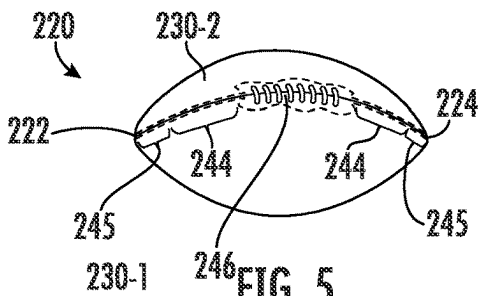


FIG. 5

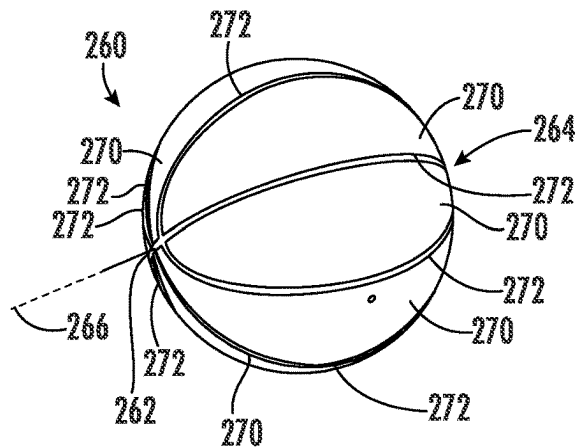


FIG. 6

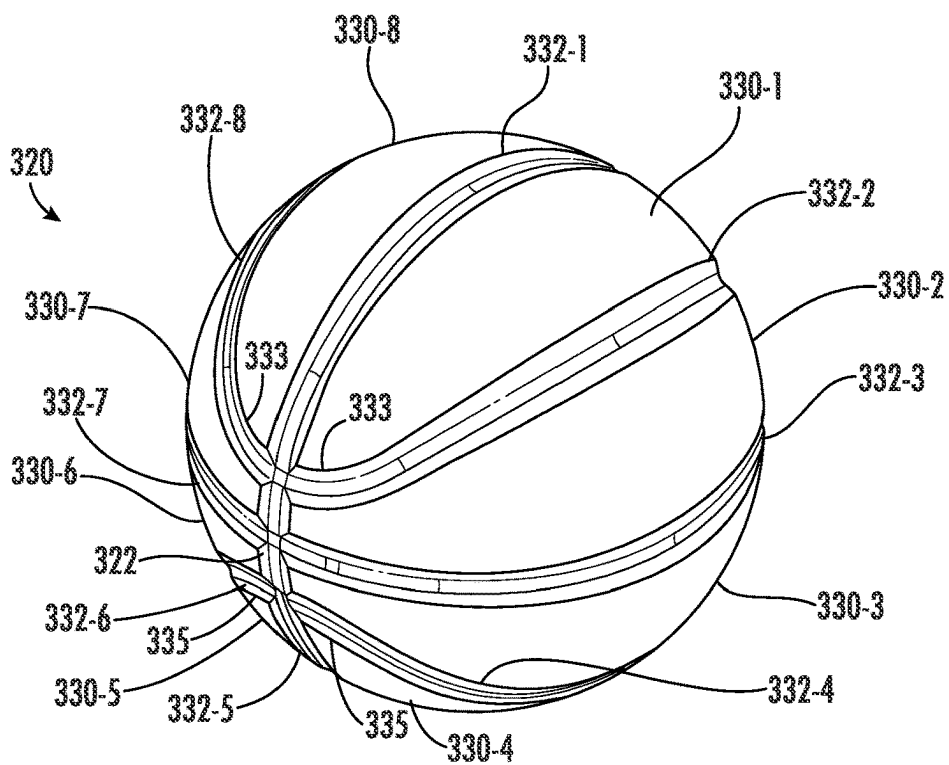


FIG. 7

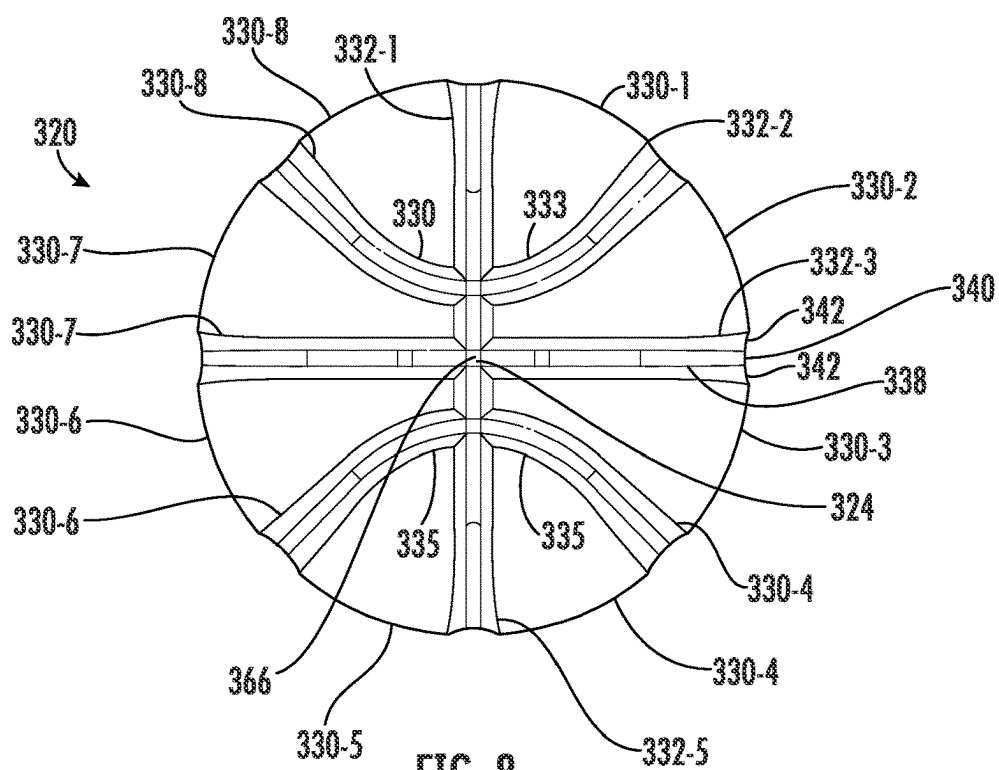


FIG. 8

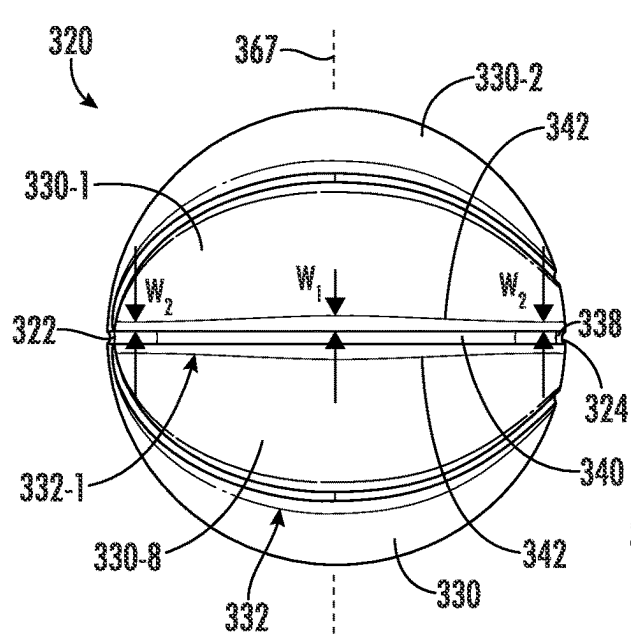


FIG. 9

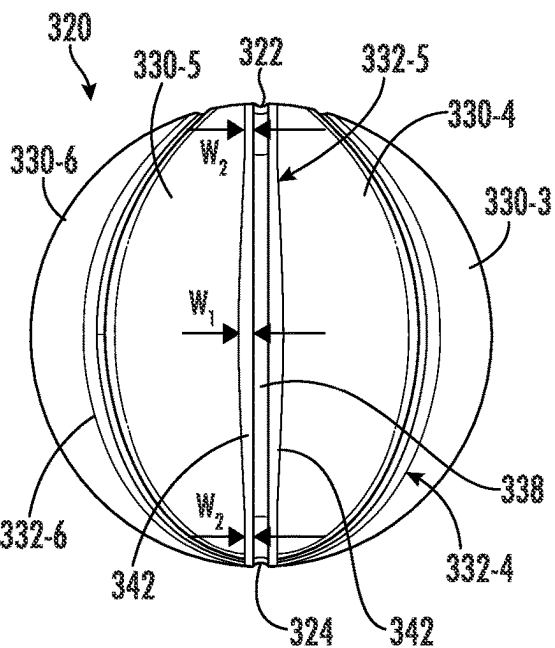


FIG. 10

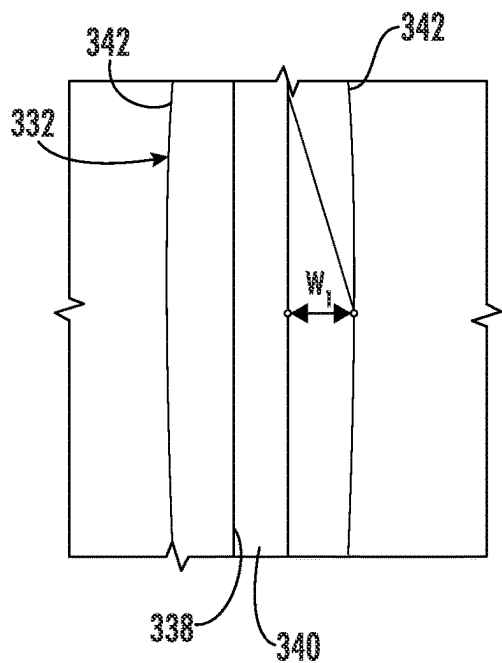


FIG. 11

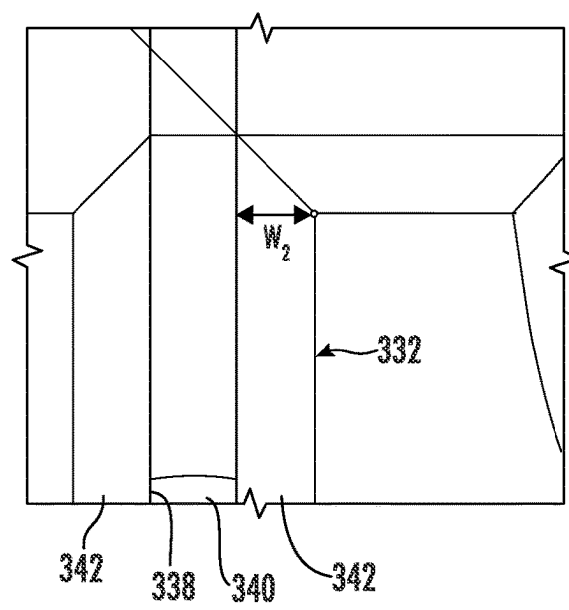
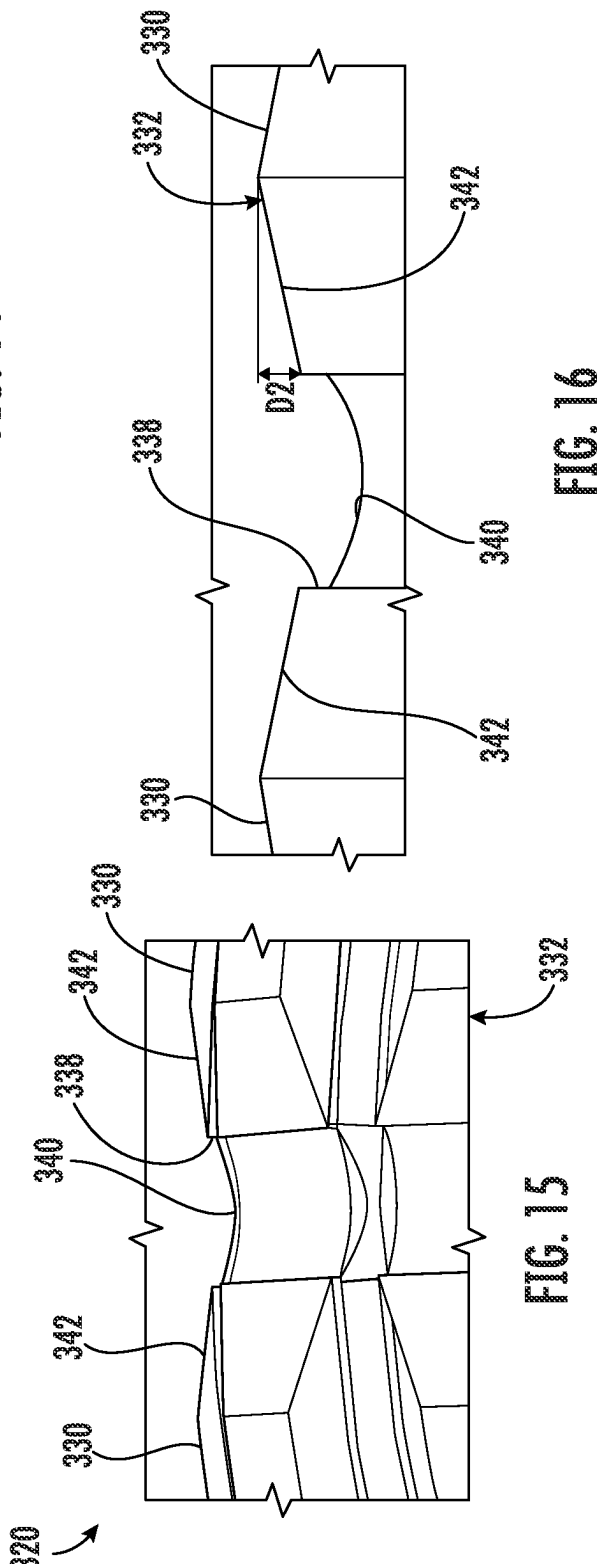
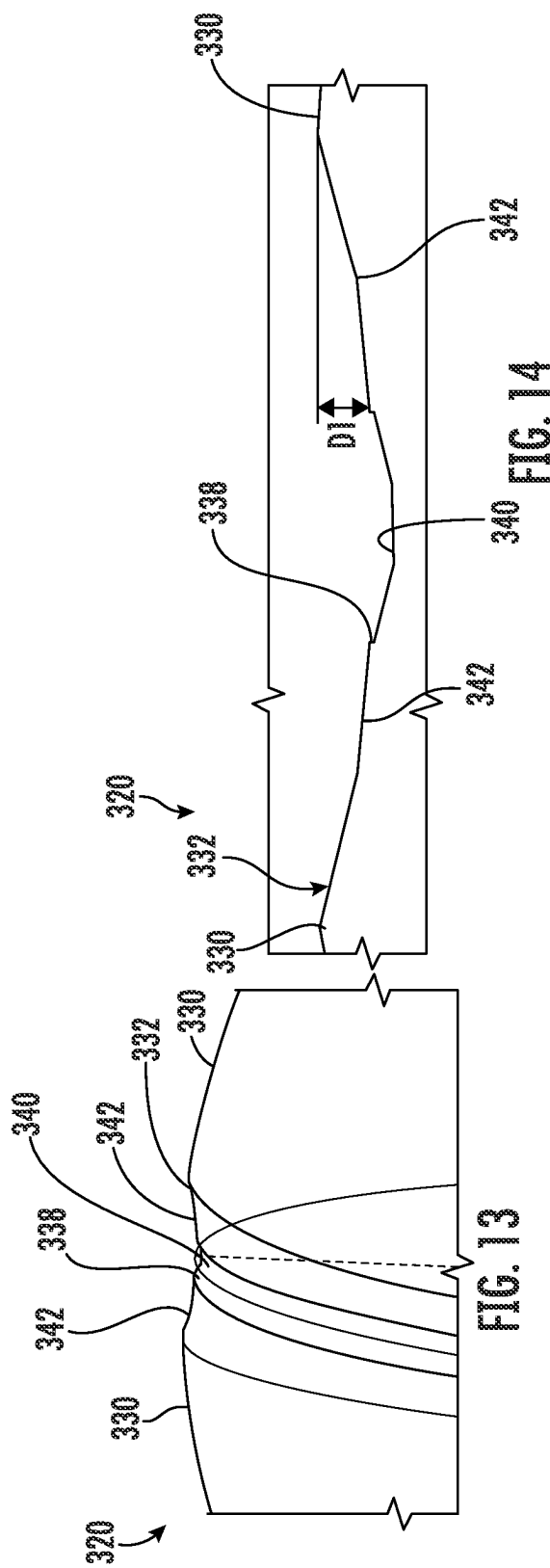


FIG. 12



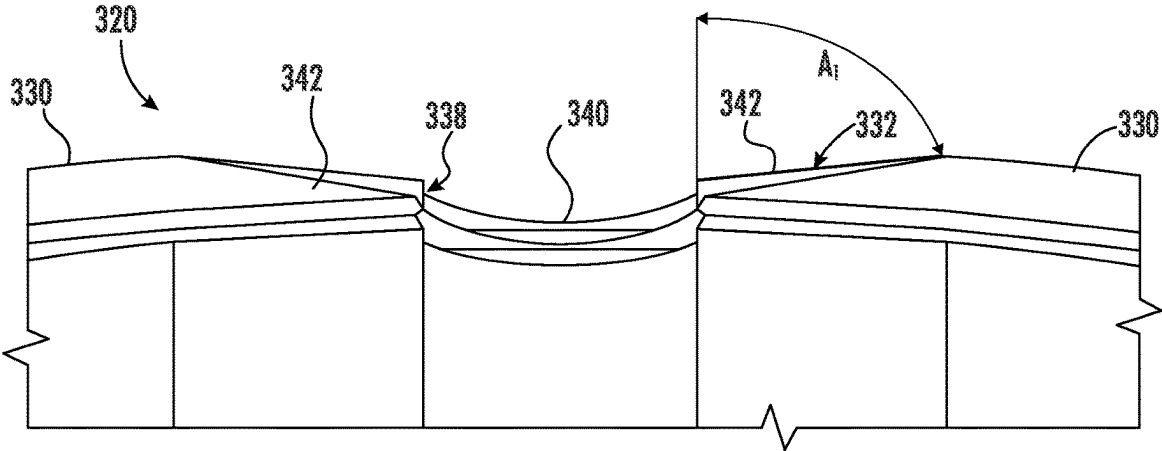


FIG. 17

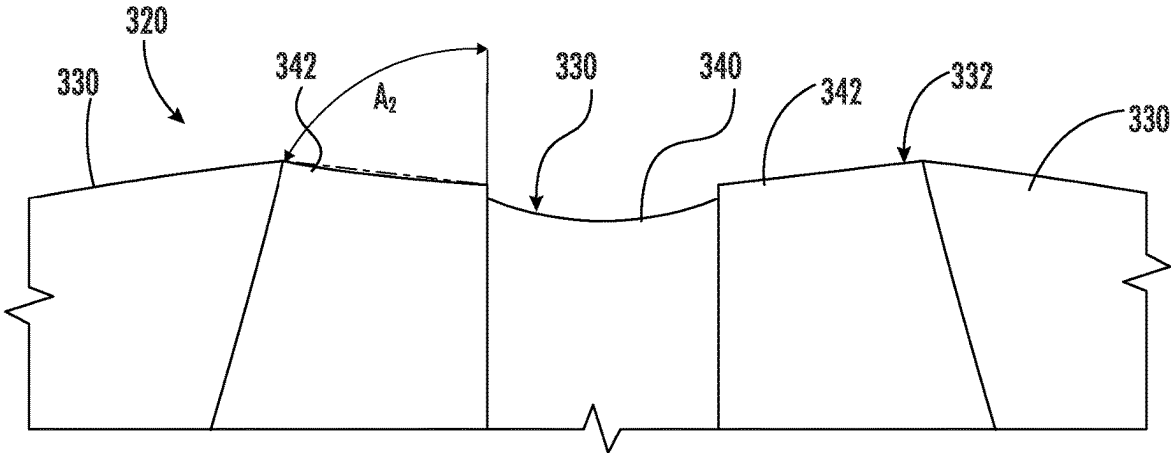


FIG. 18

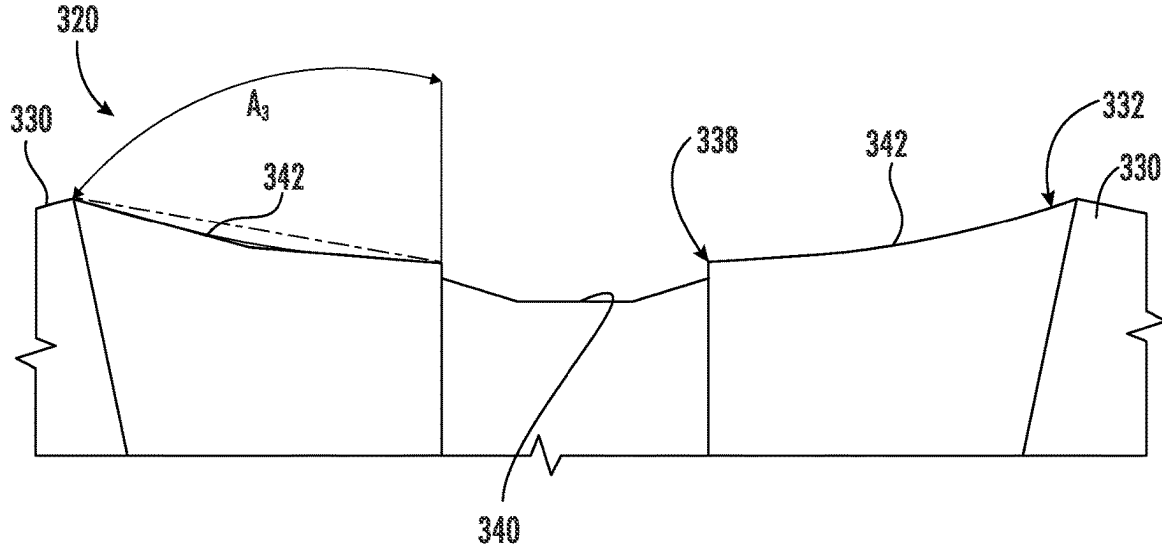
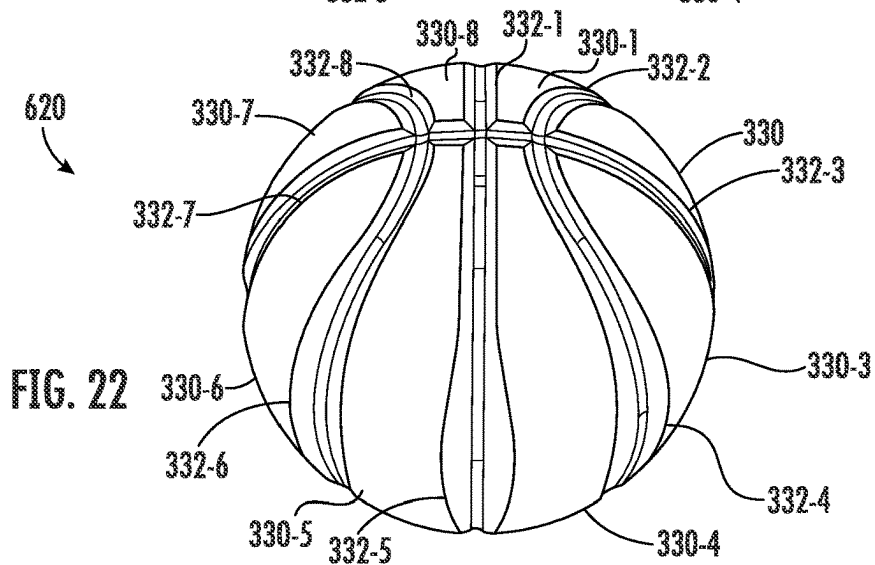
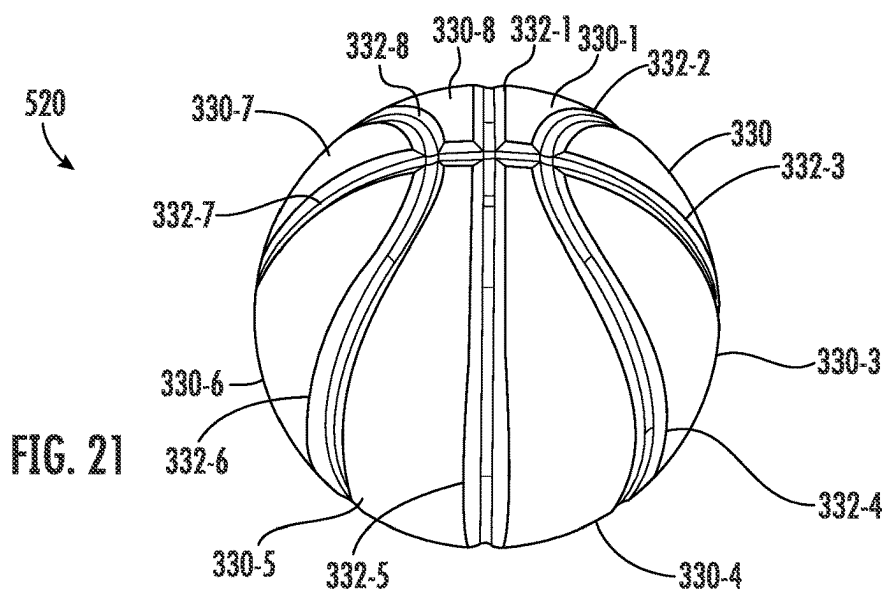
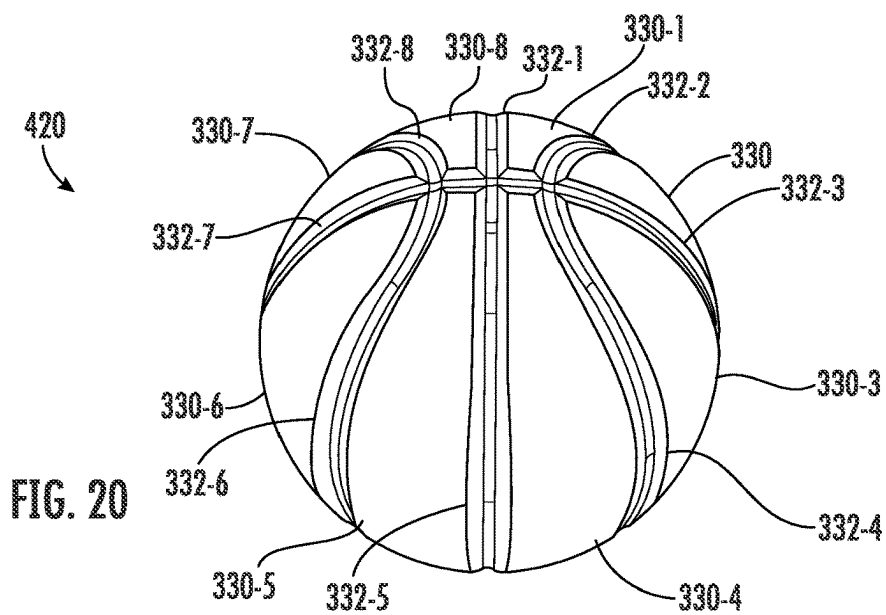


FIG. 19



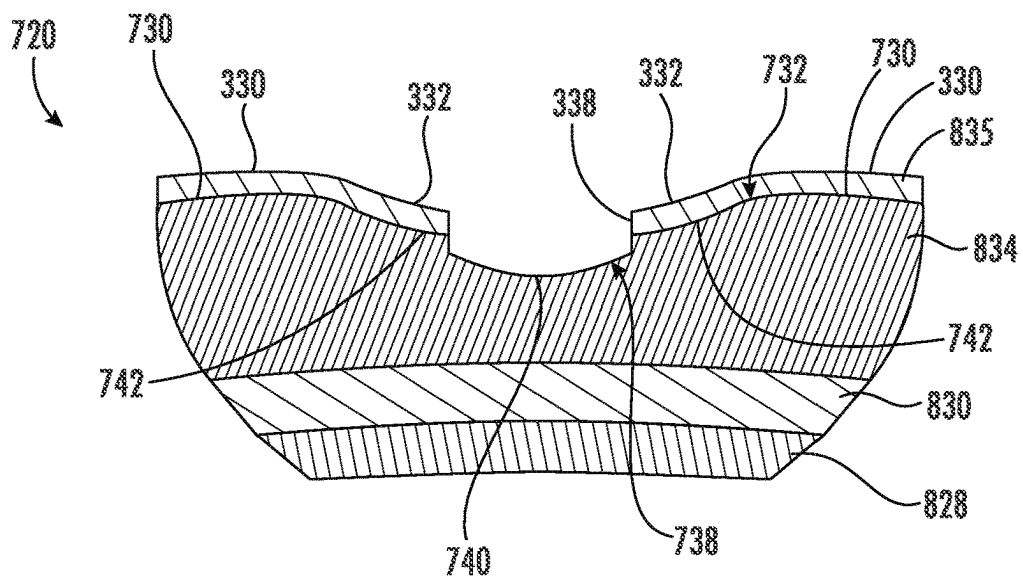


FIG. 23

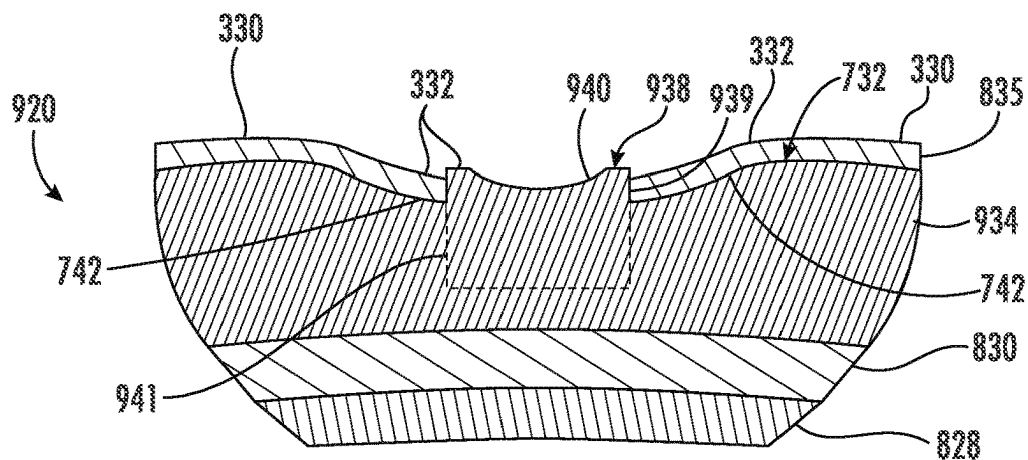


FIG. 24

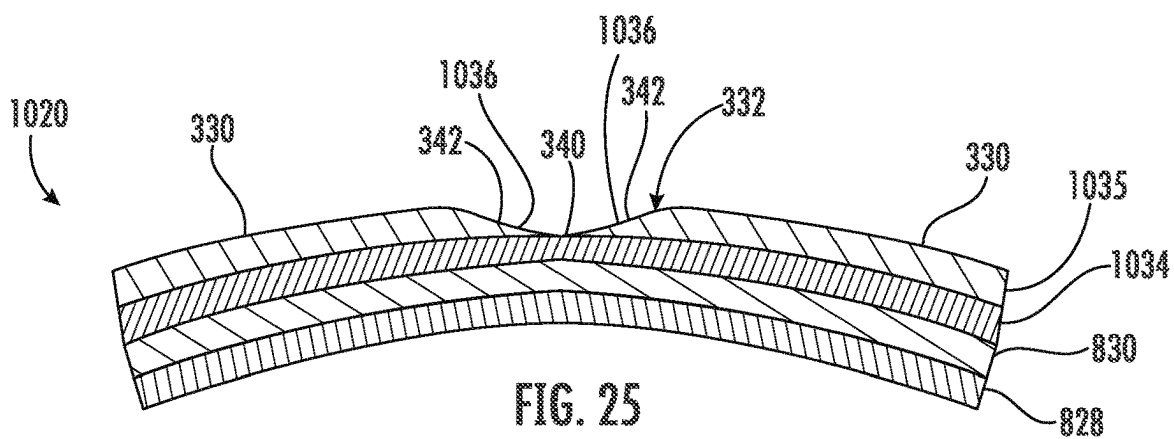


FIG. 25

GAME BALL WITH VARIABLE LOBULAR SHAPE

BACKGROUND

[0001] Game balls sometimes include an outer lobular surface formed by a series of lobes extending between opposite poles of the ball. Recessed channels separate the lobes and facilitate gripping of the game ball. One example of such a game ball is a basketball.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 is a sectional view along a first plane through an equator of an example game ball.

[0003] FIG. 2 is a sectional view along a second plane between the first plane and a pole of the example game ball of FIG. 1.

[0004] FIG. 3 is a sectional view along a first plane through an equator of an example game ball.

[0005] FIG. 4 to is a sectional view along a second plane between the first plane and a pole of the example game ball of FIG. 3.

[0006] FIG. 5 is a perspective view of an example game ball comprising a variable lobular outer surface.

[0007] FIG. 6 is a perspective view of an example game ball comprising a variable lobular outer surface.

[0008] FIG. 7 is a perspective view of an example game ball having a variable lobular outer surface.

[0009] FIG. 8 is a top view of the game ball of Figure, the bottom being a mirror image of FIG. 8.

[0010] FIG. 9 is a front view of the example game ball of FIG. 7.

[0011] FIG. 10 is a side view of the example game ball of FIG. 7.

[0012] FIG. 11 is an enlarged view of a portion of the game ball of FIG. 9 proximate to an equator of the game ball.

[0013] FIG. 12 is an enlarged view of portion of the game ball of FIG. 9 proximate to a pole of the game ball.

[0014] FIG. 13 is an enlarged view of a portion of the game ball of FIG. 9 proximate to the equator of the game ball.

[0015] FIG. 14 is a sectional view of the enlarged portion of FIG. 13.

[0016] FIG. 15 is an enlarged view of a portion of the game ball of FIG. 15 proximate to a pole of the game ball.

[0017] FIG. 16 is a sectional view of the portion of FIG. 15.

[0018] FIG. 17 is a sectional view of the game ball of FIG. 9 at a location where transitioning sides of a recessed channel on an outer surface of the game ball have their shallowest depth.

[0019] FIG. 18 is a sectional view of the game ball of FIG. 17 at a location along the recessed channel where the transitioning sides of the recessed channel begin to transition to a deeper depth.

[0020] FIG. 19 is a sectional view of the game ball of FIG. 17 at a location along the recessed channel where the transitioning sides of the recessed channel have their deepest depth.

[0021] FIG. 20 is a perspective view of an example game ball having a variable lobular outer surface.

[0022] FIG. 21 is a perspective view of an example game ball having a variable lobular outer surface.

[0023] FIG. 22 is a perspective view of an example game ball having a variable lobular outer surface.

[0024] FIG. 23 is a sectional view of portions of an example game ball having a variable lobular outer surface.

[0025] FIG. 24 is a sectional view of portions of an example game ball having a variable lobular outer surface.

[0026] FIG. 25 is a sectional view of portions of an example game ball having a variable lobular outer surface.

[0027] Throughout the drawings, identical reference numbers designate similar, but not necessarily identical, elements. The figures are not necessarily to scale, and the size of some parts may be exaggerated to more clearly illustrate the example shown. Moreover, the drawings provide examples and/or implementations consistent with the description; however, the description is not limited to the examples and/or implementations provided in the drawings.

DETAILED DESCRIPTION OF EXAMPLES

[0028] Disclosed are example game balls that have outer surfaces configured to facilitate enhanced gripping of the game ball. The outer surfaces of the example game balls comprise a first pole and a second pole opposite the first pole. The outer surfaces comprise outer circumferential portions or outer lobes that extend between the first pole and the second pole. In some implementations, the outer lobes extend from the first pole to the second pole. Recessed channels that vary in at least one of width, depth and cross-sectional shape extend along consecutive lobes.

[0029] In some implementations, each of the recessed channels comprises a floor and a pair of transitioning sides on opposite sides of the floor. The transitioning sides extend from proximate the floor outwardly to the outermost circumferential surface of the game ball. In some implementations, a surface of the floor is outwardly concave in a direction perpendicular to a centerline of the recessed channel and convex in a direction parallel to the centerline of the recessed channel. In some implementations, the surface of the floor is flat in a direction perpendicular to a centerline of the recessed channel and is convex in a direction parallel to the centerline of the recessed channel. In some implementations, the floor is inwardly pointed or has a V shape that faces outwardly.

[0030] In some implementations, at least some of the recessed channels comprises a secondary channel extending along a centerline of the recessed channel, wherein the secondary channel provides the floor or a secondary floor recessed from the floor. In some implementations, the secondary channel has a uniform depth, a uniform width or both a uniform depth and a uniform width as the secondary channel extends between the first pole and the second pole within the recessed channel. In some implementations, the floor or secondary floor provided by the secondary channel is concave in a direction perpendicular to the centerline of the secondary channel and is convex in a direction along the centerline of the secondary channel. In some implementations, the floor or secondary floor provided by the secondary channel is flat or convex in a direction perpendicular to the centerline of the secondary channel and is convex in a direction along the centerline of the secondary channel.

[0031] In some implementations, the transitioning sides are concave in a direction from the floor to opposite edges of the recessed channel. In some implementations, the transitioning sides linearly extend from the floor to the opposite edges of the recessed channel. In some implemen-

tations, the transitioning sides each have a first portion extending from the floor towards side edges of the recessed channel that is concave and a second portion extending from the side edges of the recessed channel towards a first portion that is convex.

[0032] In some implementations, the transitioning sides of the recessed channels have a first depth at a midpoint between the first pole and the second pole, and have a second depth, less than the first depth, at or proximate the first pole and at or proximate the second pole. In some implementations, the transitioning sides have a first width at a midpoint between the first pole and the second pole and have a second width, less than the first width, at or proximate the first pole and at or proximate the second pole. In such implementations, the floors extending between the transitioning sides may have a constant or uniform width or may have a width that varies.

[0033] In some implementations, the game ball (excluding the recessed channels) is spherical, such as having the shape of a basketball. In such implementations, the game ball may comprise eight lobes extending between the first pole and the second pole. In some implementations, game ball may be in the form of a prolate spheroid, wherein the first pole and the second pole are at the tips of the prolate spheroid. Examples of such a prolate spheroid include an American football or rugby ball. In such implementations, the prolate spheroid may comprise multiple lobes formed by individual panels or having the appearance of individual panels (such as in the case of a molded or 3D printed prolate spheroid), wherein the real or aesthetic panels have recessed seams that extend between the poles and that form the recessed channels, wherein the recessed channels vary in at least one of width, depth or cross-sectional shape to provide enhanced gripping of the prolate spheroid.

[0034] In some implementations, the recessed channels vary in at least one of width, depth and cross-sectional shape. In other implementations, some of the recessed channels may be uniform in width, depth and cross-sectional shape as they extend between the first pole and the second pole. One or more of the other recessed channels may vary in width, depth and/or cross-sectional shape as they extend between the first pole and the second pole. In some implementations, an individual recessed channel may vary along its length in at least one of width, depth and cross-sectional shape. In some implementations, a first one of the recessed channels may be uniform along its length and a second one of the recessed channels may also be uniform along its length, but wherein the first one of the recessed channels is different than the second one of the recessed channels in at least one of width, depth and cross-sectional shape at at least one location along its length. In some implementations, a first one of the recessed channels may vary along its length in at least one of width, depth and cross-sectional shape and a second one of the recessed channels may also vary along its length in at least one of width, depth and cross-sectional shape, where the first one of the recessed channels is also different than the second one of the recessed channels in at least one of width, depth and cross-sectional shape in corresponding locations along their lengths.

[0035] In some implementations, the varying recessed channels are formed during molding or 3D printing of the outer surface. In the case of molding, the mold forms the recessed channels that vary. In some implementations, the game ball comprises an underlying carcass that is molded,

3D printed or shaped so as to form the varying recessed channels, wherein outer cover panels of uniform thickness are placed over the carcass and cooperate with the underlying carcass to form the varying recessed channels. In some implementations, the game ball comprises an underlying carcass that does not form the varying recessed channels, wherein outer cover panels have a nonuniform thickness, where the cover panels have tapered edges that vary and are placed over the carcass to form the varying recessed channels. In some implementations, the recessed channels separate and space opposite edges of the cover panels. In some implementations, the edges of the cover panels are in abutment, wherein the edges of the cover panels form the varying recessed channels.

[0036] FIGS. 1 and 2 are sectional views illustrating portions of an example variable loby game ball 20. Game ball 20 is a “loby” game ball in that it is a lobular, having an outer surface comprising multiple lobes. For purposes of this disclosure, the term “lobular” and “loby” refer to the surface of a region having a smaller radius of curvature as compared to the overall radius of the ball surface. Game ball 20 comprises a pair of opposite poles 22, 24 joined by a longitudinal axis 26. In the example illustrated, FIG. 1 is a sectional view taken along a first radial plane perpendicular to axis 26 at an equator or midpoint of the game ball 20 between the opposite poles 22, 24. FIG. 2 is a sectional view taken along a second plane perpendicular to axis 26 at a location along axis 26 between the first plane and one of poles 22, 24.

[0037] As shown by FIGS. 1 and 2, the outer surface of game ball 20 comprises multiple outwardly facing circumferential lobes 30-1, 30-2, 30-3, 30-4, 30-5, 30-6, 30-7 and 30-8 (collectively referred to as lobes 30) that extend along axis 26 between the opposite poles 22, 24. The outer surface of game ball 20 additionally comprises recessed channels 32-1, 32-2, 32-3, 32-4, 32-5, 32-6, 32-7, and 32-8 (collectively referred to as recessed channels 32). Lobes 30 comprise outer surface protuberances or circumferential portions circumferentially extend about axis 26. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball 20.

[0038] Recessed channels 32 each extend along consecutive lobes 30 along axis 26 between poles 22, 24. In some implementations, recessed channels 32 extend, in an uninterrupted fashion from pole 22 to the other pole 24. Each of recessed channels 32 comprises a floor 40 and a pair of transitioning sides 42. Floor 40 is that portion of the recessed channel 32 that is closest to axis 26. Transitioning sides 42 extend outwardly (away from axis 26) into conjunction with the outermost circumferential surfaces of the adjacent lobes 30. In the example illustrated, each floor 40 is outwardly concave in a direction perpendicular to the centerline or length of the associated recessed channel 32 and is convex in a direction along axis 26, along the length of the associated recessed channel 32.

[0039] As shown by FIG. 1, the various recessed channels 32 vary or differ from one another. Channels 32-1 and 32-5 are similar to one another. Channels 32-2, 32-4, 32-6 and 32-8 are similar to one another. Channels 32-3 and 32-7 are similar to one another. However, recessed channels 32-1 and 32-5 are different in at least one of width, depth and cross-sectional shape as compared to recessed channels 32-2, 32-4, 32-6 and 32-7 as well as recessed channels 32-3,

32-7. Likewise, recessed channels **32-2**, **32-4**, **32-6** and **32-8** are different in at least one of width, depth and cross-sectional shape as compared to channels **32-3** and **32-7**. The variability amongst such channels **32** may provide a customized outer surface for enhanced gripping of the game ball **20**.

[0040] FIG. 2 illustrates a different portion of the outer surface of game ball **20**. As shown by FIG. 2, at the illustrated position closer to one of poles **22**, **24**, recessed channels **32-1**, **32-2**, **32-4**, **32-5**, **32-6** and **32-8** are both shallower and narrower in FIG. 2 as compared to the depth and width of the same channels at the location shown in FIG. 1. In the example illustrated, the width and depth of recessed channel **32-3** and **32-7** is uniform from the midpoint of game ball **20** shown in FIG. 1 to the pole proximate position shown in FIG. 2. In other implementations, the width and/or depth of recessed channels **32-3** and **32-7** may likewise vary along their lengths or along axis **26**.

[0041] Although game ball **20** is illustrated as comprising eight uniformly spaced lobes **30** and eight intervening recessed channels **32**, in other implementations, game ball **20** may include a fewer or greater of such lobes and a greater or fewer of such recessed channels **32**. In some implementations, the lobes and channels may be nonuniformly or non-symmetrically spaced about axis **26**. In other implementations, the relative or proportional widths and depths may be changed. For example, FIGS. 1 and 2 currently illustrate channels **32** with exaggerated sizes to better illustrate their respective depths, widths and cross-sectional shapes.

[0042] FIGS. 3 and 4 are sectional views illustrating portions of an example variable loby game ball **120**. FIGS. 3 and 4 illustrate various example shapes for the recessed channels that extend along axis **26** and along consecutive circumferential portions or lobes. As with game ball **20**, loby game ball **120** comprises a pair of opposite poles **22**, **24** joined by a longitudinal axis **26**. FIG. 3 is a sectional view taken along a first plane perpendicular to axis **26** at an equator or midpoint of the game ball **320** between the opposite poles **22**, **24**. FIG. 4 is a sectional view taken along a second plane perpendicular to axis **26** at a location along axis **26** between the first plane and one of poles **22**, **24**.

[0043] As shown by FIGS. 3 and 4, the outer surface of game ball **120** comprises multiple outwardly facing circumferential lobes **130-1**, **130-2**, **130-3**, **130-4**, **130-5**, **130-6**, **130-7** and **130-8** (collectively referred to as lobes **130**) that extend along axis **26** between the opposite poles **22**, **24**. The outer surface of game ball **120** additionally comprises recessed channels **132-1**, **132-2**, **132-3**, **132-4**, **132-5**, **132-6**, **132-7**, and **132-8** (collectively referred to as recessed channels **132**). Lobes **130** comprise outer surface protuberances or circumferential portions circumferentially extend about axis **26**. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball **120**.

[0044] In the example illustrated, recessed channels **132-1** and **132-5** have similar widths, depths and cross-sectional shapes in the plane shown by FIG. 3. Channels **132-1** and **132-5** each have an outwardly facing “V” shape with a floor **140-1** that is inwardly pointed or also has a V shape while facing outwardly. The transitioning sides **142-1** of channel **32-1** outwardly diverge from the floor. In some implementations, channels **132-1**, **132-5** may be formed by stitching to opposing panels together along axis **26** and turning the panel

inside out. In other implementations, channels **132-11** **32-5** may be 3D printed, molded or formed through material removal processes.

[0045] Recessed channels **132-2** and **132-6** have similar widths, depths and cross-sectional shapes in the plane shown by FIG. 3. Channels **132-2** and **132-6** each have a generally flat or planar floor **140-2** that is convex along axis **26**, and a pair of transitional sides **142-2** that are outwardly convex along axis **26** from the sides of floor **140** to the outer most portions of the adjacent lobes **130**.

[0046] Recessed channels **132-3** and **132-7** have similar widths, depths and cross-sectional shapes in the plane shown in FIG. 3. Channels **132-3** and **132-7** each have a generally flat or planar floor **140-3** and transitioning sides **142-3**. Transitioning sides **142-3** each have a first portion extending from the floor **140-3** towards side edges of the recessed channel **132-3**, **132-7** that is concave and a second portion extending from the side edges of the recessed channel **132-3**, **132-7** towards a first portion that is convex. As result, recessed channels **132-3**, **132-7** have an “S” shape, providing a smooth transition to the outermost surface of the adjacent lobes **130**.

[0047] As shown by FIG. 3, the various recessed channels **132** vary or differ from one another. Channels **132-1** and **132-5** are similar to one another. Channels **132-2**, and **132-6** are similar to one another. Channels **132-3** and **32-7** are similar to one another. Channels **132-4** and **132-8** are similar to one another. However, recessed channels **132-1** and **132-5** are different in at least one of width, depth and cross-sectional shape as compared to recessed channels **132-2** and **132-6** are different in at least one of width, depth and cross-sectional shape as compared to the other recessed channels of ball **120**. Likewise, recessed channels **132-3** and **132-7** are different in at least one of width, depth and cross-sectional shape as compared to the other channels of ball **120**. Recessed channels **132-4** and **132-8** are different in at least one of width, depth and cross-sectional shape as compared to other channels of ball **120**. The variability amongst such channels **32** may provide a customized outer surface for enhanced gripping of the game ball **20**.

[0048] FIG. 4 illustrates a different portion of the outer surface of game ball **120**. As shown by FIG. 2, at the illustrated position closer to one of poles **22**, **24**, each of the recessed channels **132** is shallower and narrower in FIG. 4 as compared to the depth and width of the same channels at the location shown in FIG. 3.

[0049] Although game ball **120** is illustrated as comprising eight uniformly spaced lobes **130** and eight intervening recessed channels **132**, in other implementations, game ball **120** may include a fewer or greater of such lobes and a greater or fewer of such recessed channels **132**. In some implementations, the lobes and channels may be nonuniformly or non-symmetrically spaced about axis **26**. In some implementations, game ball **120** may have a larger variety of differently shaped recessed channels or a lesser variety of differently shaped recessed channels. In some imitations, every recessed channel of game ball **120** may have one of the four different shapes shown in FIG. 3, wherein such recessed channels vary in at least one of width, depth or cross-sectional shape along their individual lengths along axis **26**. In other implementations, the relative or proportional widths and depths may be changed. For example, FIGS. 3 and 4 currently illustrate channels **132** with exag-

gerated sizes to better illustrate their respective depths, widths and cross-sectional shapes.

[0050] The lobes and intervening recessed channels shown in FIGS. 1-4 may be used in a variety of different game balls. For example, FIG. 5 illustrates an example game ball 220 in the form of a prolate spheroid, sometimes referred to as an American football or rugby ball. The outer surface of game ball 220 comprises poles 222 and 224 and comprises four lobes 230-1, 230-2 and two additional opposite lobes (not shown) extending between poles 222 to 224. Recessed channels, such as the recessed channels 140-1 described above, may be provided along consecutive or adjacent lobes 230. The recessed channels 140-1 may vary in depth and width along their length. For example, portions 244 may be provided with a recessed channel 140-1 having a first width and a first depth (similar to shown in FIG. 3) and portions 245 (proximate to poles 222 and 224) are provided with the recessed channel 140-1 having a second width and a second depth (similar to shown in FIG. 4), the second width being narrower than the first width and the second depth being shallower than the first depth. The wider and deeper portion of recessed channel 140-1 proximate to the laces 246 may provide enhanced gripping of the prolate spheroid. In other implementations, recessed channels of the game ball 220 may have other shapes and variations as described above.

[0051] FIG. 6 illustrates an example spherical loby game ball 260 in the form of a basketball. Game ball 260 comprises poles 262 and 264, lobes 270 and recessed channels 272. As with lobes 30 and 130 described above, lobes 270 comprise outer surface protuberances or circumferential portions circumferentially extend about axis 266. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball 260.

[0052] Recessed channels 272 each extend along consecutive lobes 270 along axis 266 between poles 222, 224. Each of channels 272 may be similar to any of recessed channels 32 and 132 described above. For example, in some implementations, each of channels 272 may have the shape of any one of channels 32 or 132 described above, wherein all the individual channels 272 are identical to one another along their lengths (the same cross-sectional shape, depth and width at corresponding locations along their lengths) but for their potentially differing paths along the surface of game ball 260. In such implementations, each of such channels 272 may vary along its length in at least one of width, depth and cross-sectional shape as described above with respect to channels 32 and 132 (for example, similar to how the width and depth of channel 132-8 changes from the midpoint shown in FIG. 3 to the pole proximate location shown in FIG. 4).

[0053] In some implementations, different individual channels 272 may have a uniform width, depth and cross-sectional shape along their individual lengths, but where the individual channels 272 differ amongst themselves in at least one of width, depth and cross-sectional shape at corresponding locations (for example, similar to how the width, depth and shape of channel 130-7 is different from the width, depth and shape of channel 132-8 in FIG. 3). In some implementations, a first one of channels 272 may have a first width, a first depth and a first cross-sectional shape at an equator of the game ball 260 and a second one of the channel 272 has a second width, a second depth and a second cross-sectional

shape at the equator of the game ball 260, wherein at least one of the second width, the second depth or the second cross-sectional shape differs from the first width, the first depth or the first cross-sectional shape, respectively. In some implementations, recessed channels 272 may both vary or differ amongst themselves in at least one of a width profile, a depth profile and a cross-sectional shape profile and may also individually vary or differ along their individual lengths in at least one of width, depth and cross-sectional shape. In some implementations, portions of a first individual channel 272 may be similar to corresponding portions of a second individual channel 272 while other portions of the first individual channel 272 may be different from corresponding other portions of the second individual channel 272.

[0054] FIGS. 7-10 illustrate an example spherical game ball 320 in the form of a basketball. Game ball 320 has an outer surface comprising poles 322 and 324 located along a longitudinal axis 326. Game ball 320 further comprises multiple outwardly facing circumferential lobes 330-1, 330-2, 330-3, 330-4, 330-5, 330-6, 330-7 and 330-8 (collectively referred to as lobes 330) that extend along axis 326 between the opposite poles 322, 324. The outer surface of game ball 320 additionally comprises recessed channels 332-1, 332-2, 332-3, 332-4, 332-5, 332-6, 332-7, and 332-8 (collectively referred to as recessed channels 332). Lobes 330 comprise outer surface protuberances or circumferential portions that circumferentially extend about axis 326. Although not illustrated, each of such surface portions may be pebbled or provided with other surface textures for enhanced gripping of the game ball 320.

[0055] Recessed channels each extend along consecutive lobes 330 along axis 26 between poles 322, 324. Recessed channels 332-1 and 332-5 are coplanar, extending directly opposite to one another and extending from pole 322 to pole 324. Likewise, recessed channels 332-3 and 332-7 are coplanar, extending directly opposite to one another and extending from pole 322 to pole 324. Recessed channels 332-3 and 332-7 are angularly offset from channels 332-1 and 332-5 by 90 degrees about axis 326.

[0056] Recessed channel 332-2 extends between lobes 330-1 and 330-2, equidistantly spaced from channels 332-1 and 332-3 at a midpoint or equator of game ball 320. Similarly, recessed channel 332-8 extends between lobes 330-7 and 330-7, equidistantly spaced from channels 332-1 and 332-7 at a midpoint or equator of game ball 320. Channels 332-2 and 332-8 do not extend completely to poles 322 and 324. Rather, channels 332-2 and 332-8 have opposite end portions 333 that turn towards and converge with channel 332-1 at locations in close proximity to poles 322 and 324. End portions 333 of channels 332-2 and 332-8 are aligned with one another on opposite sides of channel 332-1.

[0057] Recessed channel 332-4 extends between lobes 330-3 and 330-4, equidistantly spaced from channels 332-3 and 332-5 at a midpoint or equator of game ball 320. Similarly, recessed channel 332-6 extends between lobes 330-5 and 330-6, equidistantly spaced from channels 332-5 and 332-7 at a midpoint or equator of game ball 320. Channels 332-4 and 332-6 do not extend completely to poles 322 and 324. Rather, channels 332-4 and 332-8 have opposite end portions 335 that turn towards and converge with channel 332-5 at locations in close proximity to poles 322 and 324. End portions 335 of channels 332-4 and 332-6 are aligned with one another on opposite sides of channel 332-5.

[0058] Each of recessed channels 332 comprises a secondary channel 338 having a floor 340, and a pair of transitioning sides 342. As seen in FIGS. 14 and 16, secondary channel 338 may be further recessed from transitioning sides 342. Floor 340 may serve as the floor of each of channels 332. In the example illustrated, floor 340 is outwardly concave (inwardly curved) in a direction perpendicular to axis 326 while being convex in a direction along axis 326. In other implementations, floor 340 may be flat in a direction perpendicular to axis 326 while being convex in a direction along axis 326. In still other implementations, floor 340 may be convex in a direction perpendicular to axis 326 and convex in a direction along axis 326. In some implementations, secondary channel 338 may not be recessed from transitioning sides 342, wherein sides of floor 340 merge or converge with the lower edges of transitioning sides 342.

[0059] As shown by FIGS. 7-10, the various recessed channels 332 are each similar to one another but for the paths that they take along the surface of game ball 320. Each of channels 332 varies in at least one of depth, width and length along its length as it extends between poles 322 and 324. In the example illustrated, secondary channel 338 has a uniform width and depth along its length. It is the recessed transitioning sides 342 that vary along the length of each of channels 332 such that channels 332 vary along their lengths. In the example illustrated, each of the pair of transitioning sides on opposite sides of the second channel 338 are similar to one another, mirroring one another. In other implementations, secondary channel 338 may also vary in at least one of width, depth and cross-sectional shape along the length of the channel 332. In other implementations, the two transitioning sides on opposite sides of the same secondary channel 338 may differ from one another in at least one of width, length and cross-sectional shape.

[0060] As shown by FIGS. 11 and 12, the width of each of recessed channels 332 varies along its length. As shown by FIG. 11, at the plane 367 taken through the midpoint or equator of game ball 320, between poles 322, 324, each of the transitioning sides 342 on opposite sides of the floor 340 of the secondary channel 338 have a width W1 (also shown in FIGS. 9 and 10) in a direction perpendicular to the longitudinal axis 326 of game ball 320. As shown by FIG. 12, the same channel has a width W2 (also shown in FIGS. 9 and 10) in a direction perpendicular to the longitudinal axis 366 proximate to each of poles 322, 324. The width W-2 is smaller than the width W1. In some implementations, width W is at least 3 mm wider than width W-2. As a result, the overall width of each of channel 332 at its midpoint is at least 6 mm wider than the overall width of the same channel 332 proximate to poles 322 and 324. In some implementations, width W1 is at least 7 mm and no greater than 10 mm, and in some implementations, 8.36 mm. In such implementations, width W2 is at least 4 mm and no greater than 7 mm, and in some implementations, 5.33 mm.

[0061] As shown by FIGS. 13-16, the depth of each of the transitioning sides 342 of recessed channels 332 varies along its length. As shown by FIG. 14, at the plane 367 taken through the midpoint or equator of game ball 320, between poles 322, 324, each of the transitioning sides 342 on opposite sides of the floor 340 of the secondary channel 338 have a depth D1. As shown by FIG. 16, the same transitioning side 342 has a depth D2. Depth D1 and depth D2 preferably are sized to be within the range of 1.5 to 5.0 mm.

In some implementations, depth D2 may be smaller than depth D1. In other implementations, the depths D1 and D2 can be substantially the same. The greater depth at the midpoint or a more central portions of each of channels 332 may provide enhanced feel and enhance gripping of game ball 320. The enhance gripping may facilitate the generation of greater backspin during shooting of the game ball 320. In the example illustrated, the change in depth from the midpoint to the poles is gradual. In other implementations, the change in depth may occur in a stepwise fashion.

[0062] As further shown by FIGS. 14 and 16, the cross-sectional shape of each of channels 332 changes along its length. As shown by FIGS. 13 and 14, at the midpoint or at points in a plane taken through the equator of game ball 320, each of the transitioning sides 342 of each of channels 332 has an outwardly concave shape (in a direction perpendicular to the longitudinal axis 366) that changes to an outwardly convex shape prior to converging with the adjacent lobes 330. As shown by FIGS. 15 and 16 proximate to the poles 322, 324, each of the transitioning sides 342 of each of the channels 332 is in the shape of a linear ramp transversely extending from the circumferential surfaces of the adjacent lobes 330 to the outer sides or edges of the recessed secondary channel 338.

[0063] In the example illustrated, despite the changes in depth and width along the length of each of channels 332, transitioning from a wider and deeper sides 342 at and near the equator of the game ball to a narrower and shallower sides 342 proximate the poles of the game ball, such channels 332 are configured so as to provide a largely consistent angle along each length. The unique variable lobular shape or loby shape of the game ball of the implementations of the present invention provide additional grip points to the central regions of the game ball 320 between the poles 322 and 324. The pole regions of the game ball 320 and most game balls have high number of grip points for a player. The central region of the game ball 320 between the poles 322 and 324 offer additional grip points that facilitate a player's ability to grasp, shoot, pass, dribble and otherwise handle the game ball 320. The central region is also the preferred region of player's to hold or contact the ball when shooting, time permitting.

[0064] FIGS. 17, 18 and 19 illustrate an individual channel 332 at the shallowest location along the channel (proximate to one of poles 322, 324) for its sides 342, at the beginning of a channel drop transition along the channel, and at a deepest side depth along the channel (portions of the channel extending across the game ball equator), respectively. As shown by as shown by FIG. 17, transitioning sides 342 each have an angle A1 at the shallowest location on along the channel 332. As shown by FIG. 18, transitioning sides 342 each have an angle A2 at locations where the channel side depth begins to deepen. As shown by FIG. 19, at the deepest locations of channel sides 342, each of transitioning sides 342 have an angle A3. Angles A1, A2 and A3 are all within a range of 10 degrees, and in the example illustrated, within a range of 5 degrees. In the example illustrated, angle A1 is approximately 83.95°; angle A2 is approximately 83.99° and angle A3 is approximately 80.99 degrees. In other implementations, the angles of the transitioning sides 342 (the lobiness angle) may be less consistent along the length of each of channels 332. In some implementations, some sides 342 may have more consistent angles along their lengths than others.

[0065] FIGS. 20-22 illustrate example game balls 420, 520 and 620, respectively, in the form of basketballs. Game ball 420 is similar to game ball 320 except that each of its channels 332 are wider at the equator than the channels 332 of game ball 320. Game ball 520 is similar to game ball 420 except that each of its channels 332 are even wider still at the equator than the channels 332 of game ball 420. Game ball 620 is similar to game ball 520 except that each of its channels 332 are still wider at the equator than the channels 332 of game ball 520. As shown by FIG. 20-22, the width, depth and cross-sectional shape of channels 332 may be varied along their lengths in different manners to accommodate different playing environments.

[0066] As with the basketball shown in FIGS. 9-19, each of the basketballs shown in FIGS. 20-22 have a size, weight, and rebound consistency similar to that of standard or conventional competitive play basketballs currently sanctioned by various organizations such as the National basketball Association (NBA), National Collegiate Athletic Association (NCAA), National Federation of High Schools (NFHS) and other organizations. Each of game balls 320, 420, 520 and 620, when dropped from a height of 72" measured from the bottom of the ball, has a minimum rebound of 49" for a retail ball (measured from top of ball) and 52" for an NBA game ball. Each of such game ball 320, 420, 520 and 620 has a maximum rebound of 56 inches for an NBA ball.

[0067] FIGS. 23-25 illustrate various example game ball constructions which illustrate different ways by which the varying recess channels described above may be formed. FIGS. 23-25 illustrate different example constructions for game balls 320, 420, 520 and 620 described above. FIG. 23 is a sectional view through a portion of an example game ball 720, which may be in the form of a basketball. Game ball 720 may have an outer surface similar to the above-described outer surface of game ball 320, wherein game ball 720 comprises each of the above-described lobes 330 and recessed channels 332. Conversely, in some implementations, game ball 320 may have the cross-section shown in FIG. 23 along each of its channels 332 and may be formed as described below with respect to game ball 720.

[0068] Game ball 720 comprises inflatable bladder 828, windings 830, outer layer 834 and cover panels 835. In the example illustrated, bladder 828 may be formed primarily of rubber or a synthetic rubber. In one implementation, bladder 828 may be formed from an 80% butyl rubber and 20% natural rubber. During manufacture, the bladder may be inflated and placed in a vulcanizing or curing mold where the bladder is cured at a temperature of 160° C. After curing, the inflated bladder may be wound with a reinforcing thread to form the layer of windings 830. The layer windings may be formed from a polymer or other materials. In the example illustrated, layer of windings 830 may be formed from the approximately 2100 meters of 210 denier Nylon 66 thread. The thread may be coated with a latex or adhesive.

[0069] Outer layer 834 comprises a layer of rubber and/or synthetic rubber formed over windings 830. In some implementations, layer 834 may be formed by flat sheets or panels of rubber, such as natural and/or butyl rubber or sponge rubber (see U.S. Pat. No. 5,681,233, hereby incorporated by reference), which are laid over the wound bladder (layers 828 and 830) so as to completely cover the wound bladder. In such implementations, the inflated rubber-covered bladder is in place in a spherical carcass mold and which may be

heat molded at about 160° C. to form the carcass formed by layers 828, 830 and 834. During such molding, the rubber panels fuse to each other and into the windings 830 to form an integral outer layer 834 on the carcass.

[0070] The inside surface of the carcass mold may be shaped to form the contours of the lobular outer outside surface of game ball 720. In particular, the inside surface of the carcass mold may be shaped or configured to form each of the foundational lobes 730 and each of the recessed channels 732 in layer 834. Each of lobes 730 and recessed channels 732 may be similar to lobes 330 and channels 332, respectively, as described above and may have the layout about the game ball as shown above with respect to game ball 320. As shown by FIG. 23, each of the recessed channels formed in layer 834 comprises a secondary channel 738 having a floor 740 and a pair of transitioning sides 742 on opposite sides of floor 740. Secondary channel 738, floor 740 and sides 742 may be similar or identical to channels 338, floors 340 and sides 342 described above with respect to any of game balls 320, 420, 520 and 620. As with sides 342, sides 742 vary in depth, with cross-sectional shape along the length of each of channels 732. As with sides 342, sides 742 may be deeper and wider at an equator of the game ball and shallower and narrower proximate to the poles of the game ball.

[0071] Cover panels 835 extend over layer 834. Each of cover panels has a uniform thickness such that the outer surface contour of layer 834 is maintained. The outer surface of the completed game ball 720 has the same shape as the outer surface of layer 834. The outer surface of the completed game ball 720 comprises lobes 330, recessed channels 332, and secondary channels 338 (deeper than secondary channel 738) having the floor 740 which corresponds to floor 340. In some implementations those portions of cover panels 835 overlying sides 742 may be thinner or skived (trimmed cut away at an angle across the corner of the edges) or otherwise tapered to modify the shape, depth or width of the outermost recessed channels 332.

[0072] In some implementations, cover panels 835 may be formed from materials such as leather, synthetic leather, rubber and the like. The outer surface of each cover panel 835 may provide with a pebbled texture. Each of cover panels 835 may have a shape corresponding to the surface area of carcass which is bounded by secondary channels 738. Each of such cover panels may be adhered to layer 834 within adhesive, wherein portions of layer 34 may likewise be coated with an adhesive. In some implementations, each of cover panels 835 may have a uniform thickness of 0.031 inches (0.80 mm) throughout the panel except the variation in thickness caused by the pebbled outer surface. In other implementations, cover panel 835 may have different thicknesses within the range of 0.721.25 millimeters. In some implementations, cover panels 835 may be omitted, wherein the outer surface of layer 834 forms the outer surface of the completed game ball 720. In such implementations, the outer surface of layer 834 may be pebbled. In some implementations, windings 830 may be omitted.

[0073] FIG. 24 illustrates game ball 920, another example game ball in the form of a basketball. Game ball 920 is similar to game ball 720 except that ball 920 comprises outer layer 934 in place of outer layer 834. Those remaining components of game ball 920 which correspond to components of game ball 720 (and game ball 320) are numbered similarly.

[0074] Outer layer 934 is similar to outer layer 834 except that outer layer 934 comprises an elevated secondary channel 938 in place of the recessed secondary channel 738 in each of the channels 732. The elevated secondary channel 938 comprises a wall 939 protruding above the deepest portions of sides 742 and having a top with a floor 940. In the example illustrated, floor 940 is concave in a direction perpendicular to the longitudinal axis, such as axis 366 of the game ball and convex in a direction along the longitudinal axis of the game ball. In the example illustrated, the maximum depth of floor 940 remains above the deepest portions of sides 742. In other implementations, floor 940 may be flat in directions perpendicular to the longitudinal axis and convex in directions along the longitudinal axis. In other implementations, floor 940 may be rounded or convex in directions perpendicular to the longitudinal axis and convex in directions along the longitudinal axis. In some implementations, floor 940 may be textured. For example, floor 940 may be pebbled.

[0075] As with game ball 720, because cover panels 835 have a uniform thickness, the outer surface of the completed game ball 920 between elevated secondary channels 938 has the same shape as the outer surface of layer 934. The depth and width of the transitioning sides 742 formed in layer 934 are maintained to form transitioning sides 342 which extend between lobes 330. In the example illustrated, the thickness of cover panels 835 is less than or equal to the height of elevated channels 938; the thickness of cover panels 835 is less than or equal to the distance that elevated channels 938 protrude or rise above the deepest portions of sides 742.

[0076] In other implementations, cover panels 835 may have a greater thickness such that the outer surface of cover panels 835 extends above elevated channels 938. In such implementations, edge portions of cover panels 835 extending along an adjacent to elevated channel 938 may be tapered, skived or otherwise thinner. In some implementations, cover panels 835 may be omitted from game ball 920. In such implementations, the outer surface of layer 934 forming lobes 730, sides 742 and floor 940 may be pebbled for enhanced grip. In some implementations, as shown by broken lines 941, rather than being integrally formed as a single unitary body as part of layer 934, the elevated channel 938 may comprise a separate strip of material that fused or bonded to a top of layer 934 or that is inset into a groove formed in layer 934 prior to being fused or bonded in place.

[0077] FIG. 25 illustrates game ball 1020, another example game ball in the form of a basketball. Game ball 1020 is similar to game ball 720 except that ball 920 comprises outer layer 1034 and cover panels 1035 in place of outer layer 834 and cover panels 835, respectively. Those remaining components of game ball 920 which correspond to components of game ball 720 (and game ball 320) are numbered similarly.

[0078] Outer layer 1034 is similar to outer layer 834 except that outer layer 1034 does not include sides corresponding to sides 342 of the recessed channels 332. In contrast, outer layer 1034 may have a uniform thickness about the entire circumference of the game ball on 1020. Portions of the outer layer 1034 form floors 340 of each of the recessed channels 332. Cover panels 1035 are similar to cover panels 835 described above except that edge portions 1036 of cover panel 1035 are shaped, thinned, tapered, cut-away or skived so as to form the above-described

transitioning sides 342 along the opposite sides of each of above-described channels 332.

[0079] In some implementations, those portions of layer 1034 serving as a floors 340 of the channels 332 may have a different thickness than remaining portions of the layer 1034. For example, such portions may be concave in directions perpendicular to the longitudinal axis of game ball or convex in directions perpendicular to the longitudinal axis of the game ball (transversely across channel 332). Such portions may be pebbled. As described above with respect to FIG. 24, in some implementations, layer 1034 may include an integrally formed elevated channel 938 between the transitioning sides 342 provided by the edges of panel 1035. In some implementations, as shown by broken lines 941 in FIG. 24, and elevated channel 938 in the form of strip or multiple strips may be fused or bonded to a top of layer 1034 or inset within a groove or channel formed within layer 1034 between the tapered, cut-away or skived edges of panel 1035.

[0080] Although the present disclosure has been described with reference to example implementations, workers skilled in the art will recognize that changes may be made in form and detail without departing from the scope of the claimed subject matter. For example, although different example implementations may have been described as including features providing benefits, it is contemplated that the described features may be interchanged with one another or alternatively be combined with one another in the described example implementations or in other alternative implementations. Because the technology of the present disclosure is relatively complex, not all changes in the technology are foreseeable. The present disclosure described with reference to the example implementations and set forth in the following claims is manifestly intended to be as broad as possible. For example, unless specifically otherwise noted, the claims reciting a single particular element also encompass a plurality of such particular elements. The terms “first”, “second”, “third” and so on in the claims merely distinguish different elements and, unless otherwise stated, are not to be specifically associated with a particular order or particular numbering of elements in the disclosure.

What is claimed is:

1. A variable loby game ball comprising:
an outer surface comprising:
a first pole and a second pole opposite the first pole;
outer lobes extending between the first pole and the second pole, herein recessed channels that vary in at least one of width, depth and cross-sectional shape extend along consecutive lobes.
2. The variable loby game ball of claim 1, wherein the recessed channels each comprise a floor and a pair of transitioning sides, wherein the transitioning sides vary in at least one of width, depth and cross-sectional shape.
3. The variable loby game ball of claim 2 wherein each of the recessed channels has a secondary channel forming the floor between the pair of transitioning sides.
4. The variable loby game ball of claim 2, wherein the transitioning sides extend from proximate the first pole to proximate the second pole and wherein the transitioning sides vary in at least one of depth and width along their lengths between the first pole and the second pole.
5. The variable loby game ball of claim 4, wherein the transitioning sides vary in both depth and width along their lengths between the first pole and the second pole.

6. The variable loby game ball of claim 4, wherein the transitioning sides vary in width along their lengths between the first pole and the second pole.

7. The variable loby game ball of claim 6, wherein the transitioning sides have a first width proximate the first pole and the second pole and a second width, greater than the first width, at a midpoint between the first pole and the second pole.

8. The variable loby game ball of claim 7, wherein the transitioning sides have a first depth proximate the first pole and the second pole and a second depth, greater than the first depth, at the midpoint between the first pole and the second pole.

9. The variable loby game ball of claim 8, wherein the pairs of transitioning sides proximate the first pole and the second pole extend at a first angle towards the channel and wherein the pairs of transitioning sides at the midpoint between the first pole and the second pole extend towards the channel at a second angle, the second angle being within 5 degrees of the first angle.

10. The variable loby game ball of claim 7, wherein the transitioning sides are concave at the midpoint and convex proximate the first pole and the second pole.

11. The variable loby game ball of claim 7, wherein the transitioning sides extend an angle of at least 75 degrees and no greater than 88 degrees relative to a radial plane passing through a center of the variable loby game ball.

12. The variable loby game ball of claim 4, wherein the transitioning sides extending along the opposite sides of each channel and mirror one another.

13. The variable loby game ball of claim 2, wherein the recessed channels comprise a first channel having a first pair of transitioning sides and a second channel having a second

pair of transitioning sides different with respect to at least one of width, depth and cross-sectional shape of the first pair of transitioning sides.

14. The variable loby game ball of claim 13, wherein the transitioning sides extend from proximate the first pole to proximate the second pole and wherein the transitioning sides vary in at least one of depth and width along their lengths between the first pole and the second pole.

15. The variable loby game ball of claim 14, wherein the transitioning sides vary in both depth and width along their lengths between the first pole and the second pole.

16. The variable loby game ball of claim 14, wherein the transitioning sides have a first width proximate the first pole and the second pole and a second width, greater than the first width, at a midpoint between the first pole and the second pole.

17. The variable loby game ball of claim 16, wherein the transitioning sides are concave at the midpoint and convex proximate the first pole and the second pole.

18. The variable loby game ball of claim 2 further comprising:

a carcass; and

a plurality of cover panels secured to the carcass.

19. The variable loby game ball of claim 18, wherein the carcass has a lobular outer carcass surface having foundational lobes corresponding to the lobes of the outer surface.

20. The variable loby game ball of claim 19, wherein the plurality of cover panels have a non-uniform thickness so as to form the pair of transitioning sides along the floor.

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